

Tests

- Baseline and follow-up liver function tests

Side effects

How Drug Causes Side Effects

- When alcohol is consumed by a patient taking disulfiram, levels of acetaldehyde build up, causing side effects of alcohol toxicity
- One of disulfiram's metabolites is carbon disulfide, which may be excreted through the lungs; this could account for the side effect of metallic taste

Notable Side Effects

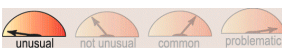
- Metallic taste, dermatitis, sedation
- Flushing, headache, tachycardia, nausea, vomiting (if alcohol is consumed)



Life-Threatening or Dangerous Side Effects

- Hepatotoxicity
- Myocardial infarction, congestive heart failure, respiratory depression, other signs of alcohol toxicity (if alcohol is consumed)

Weight Gain



- Reported but not expected

Sedation



- Occurs in significant minority

What to Do About Side Effects

- Wait
- Reduce dose
- Take at night to reduce sedation

Best Augmenting Agents for Side Effects

- Dose reduction or switching to another agent may be more effective since most side effects cannot be improved with an augmenting agent

Dosing and use

Usual Dosage Range

- 250–500 mg/day; 1-year duration

Dosage Forms

- Tablet 250 mg, 500 mg scored

How to Dose

- The patient should not take disulfiram until at least 12 hours after drinking
- Initial 250–500 mg/day for 1–2 weeks
- Usually dosed in the morning but can be dosed at night if sedation is a problem
- Maintenance dose usually 250 mg/day; maximum dose 500 mg/day



Dosing Tips

- The patient must be fully informed of the disulfiram-alcohol reaction
- The patient should be advised not to consume any food or beverages containing alcohol or to use any alcohol-containing preparations (e.g., cough syrup)
- The patient should be warned that reactions may occur up to 2 weeks after disulfiram is stopped
- The patient should carry an emergency card stating that he or she is taking disulfiram

Overdose

- Unknown

Long-Term Use

- Maintenance treatment should be continued until the patient is recovered

Habit Forming

- No

How to Stop

- Taper not necessary
- A disulfiram-alcohol reaction may occur for up to 2 weeks after disulfiram is stopped

Pharmacokinetics

- Half-life of parent drug is 60–120 hours
- Half-life of metabolites is 13.9 hours (diethyldithiocarbamate) and 8.9 hours (carbon disulfide)



Drug Interactions

- Disulfiram may increase blood levels of phenytoin; baseline and follow-up levels of phenytoin should be taken
- Disulfiram may prolong prothrombin time, requiring dose adjustment of oral anticoagulants
- Use with isoniazid may lead to unsteady gait or change in mental status

Other Warnings/Precautions

- Disulfiram should not be given to a patient in a state of alcohol intoxication or without the patient's full knowledge
- Not recommended for patients older than age 60 or for those with severe pulmonary disease, chronic renal failure, diabetes, peripheral neuropathy, seizures, cirrhosis, or portal hypertension
- Use with extreme caution in patients with hypothyroidism, epilepsy, cerebral damage
- Patients taking disulfiram should not be exposed to ethylene dibromide or its vapors, as this has resulted in a higher incidence of tumors in rats

Do Not Use

- If the patient is in a state of alcohol intoxication
- Without the patient's full knowledge
- For at least 12 hours after the patient last drank
- If patient is taking metronidazole, amprenavir, ritonavir, or sertraline
- If patient has psychosis
- If patient has cardiovascular disease
- If there is a proven allergy to disulfiram
- If there is a proven allergy to thiuram derivatives

Special populations

Renal Impairment

- Not recommended for patients with chronic renal failure

Hepatic Impairment

- Not recommended

Cardiac Impairment

- Contraindicated

Elderly

- Not generally recommended for patients older than age 60
- Some patients may tolerate lower doses better



Children and Adolescents

- Safety and efficacy have not been established



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001

- Controlled studies have not been conducted in pregnant women
- Some animal studies have shown adverse effects
- Pregnant women needing to stop drinking may consider behavioral therapy before pharmacotherapy
- Not generally recommended for use during pregnancy, especially during first trimester

Breast Feeding

- Unknown if disulfiram is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed

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Potential Advantages

- Individuals who are motivated to abstain from alcohol

Potential Disadvantages

- Adherence rates can be low

Primary Target Symptoms

- Alcohol dependence

- Some evidence of efficacy in comorbid alcohol use disorder and PTSD
- Preliminary evidence of efficacy for use in cocaine dependence, both alone and comorbid with alcohol use disorder

Suggested Reading

Barth KS , Malcolm RJ . Disulfiram: an old therapeutic with new applications . *CNS Neurol Disord Drug Targets* 2010 ;**9** (1):5 –12.

Jorgensen CH , Pedersen B , Tonnesen H. The efficacy of disulfiram for the treatment of alcohol use disorder . *Alcohol Clin Exp Res* 2011 ;**35** (10):1749 –58.

Pani PP , Troqu E , Vacca R , et al. Disulfiram for the treatment of cocaine dependence . *Cochrane Database Syst Rev* 2010 ;**20** (1):CD007024 .

Donepezil

Therapeutics

Brands

- Aricept
- Memac
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: acetylcholine enzyme inhibitor (ACh-EI)
- Cholinesterase inhibitor (selective acetylcholinesterase inhibitor); cognitive enhancer

Commonly Prescribed for

(bold for FDA approved)

- **Alzheimer disease (mild, moderate, and severe)**
- Memory disorders in other conditions
- Mild cognitive impairment

How the Drug Works

- * Reversibly but noncompetitively inhibits centrally active acetylcholinesterase (AChE), making more acetylcholine available
- Increased availability of acetylcholine compensates in part for degenerating cholinergic neurons in neocortex that regulate memory
- Does not inhibit butyrylcholinesterase
- May release growth factors or interfere with amyloid deposition

How Long Until It Works

- May take up to 6 weeks before any improvement in baseline memory or behavior is evident
- May take months before any stabilization in degenerative course is evident

If It Works

- May improve symptoms and slow progression of disease, but does not reverse the degenerative process

If It Doesn't Work

- Consider adjusting dose, switching to a different cholinesterase inhibitor, or adding an appropriate augmenting agent
- Reconsider diagnosis and rule out other conditions such as depression or a dementia other than Alzheimer disease

Best Augmenting Combos for Partial Response or Treatment Resistance

- * Atypical antipsychotics to reduce behavioral disturbances
- * Antidepressants if concomitant depression, apathy, or lack of interest
- * Memantine for moderate to severe Alzheimer disease
- Divalproex, carbamazepine, or oxcarbazepine for behavioral disturbances
- Not rational to combine with another cholinesterase inhibitor

Tests

- None for healthy individuals

Side effects

How Drug Causes Side Effects

- Peripheral inhibition of acetylcholinesterase can cause gastrointestinal side effects
- Central inhibition of acetylcholinesterase may contribute to nausea, vomiting, weight loss, and sleep disturbances

Notable Side Effects

- * Nausea, diarrhea, vomiting, appetite loss, increased gastric acid secretion, weight loss

- Insomnia, dizziness
- Muscle cramps, fatigue, depression, abnormal dreams



Life-Threatening or Dangerous Side Effects

- Rare seizures
- Rare syncope

Weight Gain



- Reported but not expected
- Some patients may experience weight loss

Sedation



- Reported but not expected

What to Do About Side Effects

- Wait
- Wait
- Wait
- Take in daytime to reduce insomnia
- Use slower dose titration

- Consider lowering dose, switching to a different agent, or adding an appropriate augmenting agent

Best Augmenting Agents for Side Effects

- Hypnotics or trazodone may improve insomnia
- Many side effects cannot be improved with an augmenting agent

Dosing and use

Usual Dosage Range

- 5–10 mg at night

Dosage Forms

- Tablet 5 mg, 10 mg, 23 mg
- Orally disintegrating tablet 5 mg, 10 mg
- Namzaric extended-release (combination memantine/donepezil) 7 mg/10 mg, 14 mg/10 mg, 21 mg/10 mg, 28 mg/10 mg

How to Dose

- Initial 5 mg/day; may increase to 10 mg/day after 4–6 weeks
- Namzaric, for patients already on donepezil 10 mg once daily: initial dose 7 mg/10 mg once daily in the evening; increase weekly in 7 mg increments; maximum dose 28 mg/10 mg

- Namzaric, for patients already on memantine (10 mg twice per day or 28 mg extended-release once per day) and donepezil 10 mg once daily: can switch to 28 mg/10 mg once daily in the evening



Dosing Tips

- Side effects occur more frequently at higher doses than at lower doses
- Slower titration (e.g., 6 weeks to 10 mg/day) may reduce the risk of side effects
- Food does not affect the absorption of donepezil
- Probably best to utilize highest tolerated dose within the usual dosage range
- Some off-label uses for cognitive disturbances other than Alzheimer disease have anecdotally utilized doses higher than 10 mg/day
- * When switching to another cholinesterase inhibitor, probably best to cross-titrate from one to the other to prevent precipitous decline in function if the patient washes out of one drug entirely
- Namzaric can be taken with or without food, whole or sprinkled on applesauce; do not divide, chew, or crush

Overdose

- Can be lethal; nausea, vomiting, excess salivation, sweating, hypotension, bradycardia, collapse, convulsions, muscle weakness (weakness of respiratory muscles can lead to death)

Long-Term Use

- Drug may lose effectiveness in slowing degenerative course of Alzheimer disease after 6 months
- Can be effective in some patients for several years

Habit Forming

- No

How to Stop

- Taper to avoid withdrawal effects
- Discontinuation may lead to notable deterioration in memory and behavior, which may not be restored when drug is restarted or another cholinesterase inhibitor is initiated

Pharmacokinetics

- Metabolized by CYP450 2D6 and CYP450 3A4
- Elimination half-life approximately 70 hours



Drug Interactions

- Donepezil may increase the effects of anesthetics and should be discontinued prior to surgery
- Inhibitors of CYP450 2D6 and CYP450 3A4 may inhibit donepezil metabolism and increase its plasma levels

- Inducers of CYP450 2D6 and CYP450 3A4 may increase clearance of donepezil and decrease its plasma levels
- Donepezil may interact with anticholinergic agents and the combination may decrease the efficacy of both
- May have synergistic effect if administered with cholinomimetics (e.g., bethanechol)
- Bradycardia may occur if combined with beta blockers
- Theoretically, could reduce the efficacy of levodopa in Parkinson's disease
- Not rational to combine with another cholinesterase inhibitor

Other Warnings/Precautions

- May exacerbate asthma or other pulmonary disease
- Increased gastric acid secretion may increase the risk of ulcers
- Bradycardia or heart block may occur in patients with or without cardiac impairment

Do Not Use

- If there is a proven allergy to donepezil

Special populations

Renal Impairment

- Few data available but dose adjustment is most likely unnecessary
- Severe renal impairment: the recommended maintenance dose for Namzaric is 14 mg/10 mg once daily in the evening

Hepatic Impairment

- Few data available; may need to lower dose

Cardiac Impairment

- Should be used with caution
- Syncopal episodes have been reported with the use of donepezil

Elderly

- Some patients may tolerate lower doses better
- Use of cholinesterase inhibitors may be associated with increased rates of syncope, bradycardia, pacemaker insertion, and hip fracture in older adults with dementia

Children and Adolescents

- Safety and efficacy have not been established
- Preliminary reports of efficacy as an adjunct in attention deficit hyperactivity disorder (ADHD) (ages 8–17)

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- * Not recommended for use in pregnant women or women of childbearing potential

Breast Feeding

- Unknown if donepezil is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed
- Donepezil is not recommended for use in nursing women

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Potential Advantages

- Once a day dosing
- May be used in vascular dementia
- May work in some patients who do not respond to other cholinesterase inhibitors

- May work in some patients who do not tolerate other cholinesterase inhibitors

Potential Disadvantages

- Patients with insomnia

Primary Target Symptoms

- Memory loss in Alzheimer disease
- Behavioral symptoms in Alzheimer disease
- Memory loss in other dementias

Pearls

- A fixed-dose combination of memantine extended-release and donepezil (Namzaric) has been approved for the treatment of moderate to severe Alzheimer dementia in patients stabilized on memantine and donepezil
 - Dramatic reversal of symptoms of Alzheimer disease is not generally seen with cholinesterase inhibitors
 - Can lead to therapeutic nihilism among prescribers and lack of an appropriate trial of a cholinesterase inhibitor
- * Perhaps only 50% of Alzheimer patients are diagnosed, and only 50% of those diagnosed are treated, and only 50% of those treated are given a cholinesterase inhibitor, and then only for 200 days in a disease that lasts 7–10 years

- Must evaluate lack of efficacy and loss of efficacy over months, not weeks
- * Treats behavioral and psychological symptoms of Alzheimer dementia as well as cognitive symptoms (i.e., especially apathy, disinhibition, delusions, anxiety, cooperation, pacing)
- Patients who complain themselves of memory problems may have depression, whereas patients whose spouses or children complain of the patient's memory problems may have Alzheimer disease
- Treat the patient but ask the caregiver about efficacy
- What you see may depend upon how early you treat
- The first symptoms of Alzheimer disease are generally mood changes; thus, Alzheimer disease may initially be diagnosed as depression
- Women may experience cognitive symptoms in perimenopause as a result of hormonal changes that are not a sign of dementia or Alzheimer disease
- Aggressively treat concomitant symptoms with augmentation (e.g., atypical antipsychotics for agitation, antidepressants for depression)
- If treatment with antidepressants fails to improve apathy and depressed mood in the elderly, it is possible that this represents early Alzheimer disease and a cholinesterase inhibitor like donepezil may be helpful
- What to expect from a cholinesterase inhibitor:

- Patients do not generally improve dramatically although this can be observed in a significant minority of patients
- Onset of behavioral problems and nursing home placement can be delayed
- Functional outcomes, including activities of daily living, can be preserved
- Caregiver burden and stress can be reduced
- Delay in progression in Alzheimer disease is not evidence of disease-modifying actions of cholinesterase inhibition
- Cholinesterase inhibitors like donepezil depend upon the presence of intact targets for acetylcholine for maximum effectiveness and thus may be most effective in the early stages of Alzheimer disease
- The most prominent side effects of donepezil are gastrointestinal effects, which are usually mild and transient
- * May cause more sleep disturbances than some other cholinesterase inhibitors
- For patients with intolerable side effects, generally allow a washout period with resolution of side effects prior to switching to another cholinesterase inhibitor
- Weight loss can be a problem in Alzheimer patients with debilitation and muscle wasting
- Women over 85, particularly with low body weights, may experience more adverse effects

- Use with caution in underweight or frail patients
 - Cognitive improvement may be linked to substantial (>65%) inhibition of acetylcholinesterase
 - Donepezil has greater action on CNS acetylcholinesterase than on peripheral acetylcholinesterase
 - Some Alzheimer patients who fail to respond to donepezil may respond to another cholinesterase inhibitor
 - Some Alzheimer patients who fail to respond to another cholinesterase inhibitor may respond when switched to donepezil
 - To prevent potential clinical deterioration, generally switch from long-term treatment with one cholinesterase inhibitor to another without a washout period
- * Donepezil may slow the progression of mild cognitive impairment to Alzheimer disease
- * May be useful for dementia with Lewy bodies (DLB, constituted by early loss of attentiveness and visual perception with possible hallucinations, Parkinson-like movement problems, fluctuating cognition such as daytime drowsiness and lethargy, staring into space for long periods, episodes of disorganized speech)
- May decrease delusions, apathy, agitation, and hallucinations in dementia with Lewy bodies
- * May be useful for vascular dementia (e.g., acute onset with slow stepwise progression that has plateaus, often with gait abnormalities, focal signs, imbalance, and urinary incontinence)

- May be helpful for dementia in Down's syndrome
- Suggestions of utility in some cases of treatment-resistant bipolar disorder
- Theoretically, may be useful for ADHD, but not yet proven
- Theoretically, could be useful in any memory condition characterized by cholinergic deficiency (e.g., some cases of brain injury, cancer chemotherapy-induced cognitive changes, etc.)

Suggested Reading

Bentue-Ferrer D , Tribut O , Polard E , Allain H . Clinically significant drug interactions with cholinesterase inhibitors: a guide for neurologists . *CNS Drugs* 2003 ;**17** :947–63 .

Birks JS , Harvey R . Donepezil for dementia due to Alzheimer's disease . *Cochrane Database Syst Rev* 2003 ;(1):CD001190 .

Bonner LT , Peskind ER . Pharmacologic treatments of dementia . *Med Clin North Am* 2002 ;**86** :657–74 .

Jones RW . Have cholinergic therapies reached their clinical boundary in Alzheimer's disease ? *Int J Geriatr Psychiatry* 2003 ;**18** (Suppl 1):S7 –13.

Seltzer B . Donepezil: an update . *Expert Opin Pharmacother* 2007 ;**8** (7):1011 –23.

Dothiepin

Therapeutics

Brands

- Prothiaden
- *see index for additional brand names*

Generic?

In UK

Class

- Neuroscience-based Nomenclature: serotonin, norepinephrine multimodal (SN-MM)
- Tricyclic antidepressant (TCA)
- Serotonin and norepinephrine/noradrenaline reuptake inhibitor

Commonly Prescribed for

(bold for FDA approved)

- Major depressive disorder
- Anxiety
- Insomnia
- Neuropathic pain/chronic pain

- Treatment-resistant depression

How the Drug Works

- Boosts neurotransmitters serotonin and norepinephrine/noradrenaline
- Blocks serotonin reuptake pump (serotonin transporter), presumably increasing serotonergic neurotransmission
- Blocks norepinephrine reuptake pump (norepinephrine transporter), presumably increasing noradrenergic neurotransmission
- Presumably desensitizes both serotonin 1A receptors and beta adrenergic receptors
- Since dopamine is inactivated by norepinephrine reuptake in frontal cortex, which largely lacks dopamine transporters, dothiepin can increase dopamine neurotransmission in this part of the brain

How Long Until It Works

- May have immediate effects in treating insomnia or anxiety
- Onset of therapeutic actions usually not immediate, but often delayed 2–4 weeks
- If it is not working within 6–8 weeks for depression, it may require a dosage increase or it may not work at all
- May continue to work for many years to prevent relapse of symptoms